

Frequently Asked Questions for xD1000 VFD

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1. How to do basic programming of xD1000 to give Start command & frequency reference from Terminals?

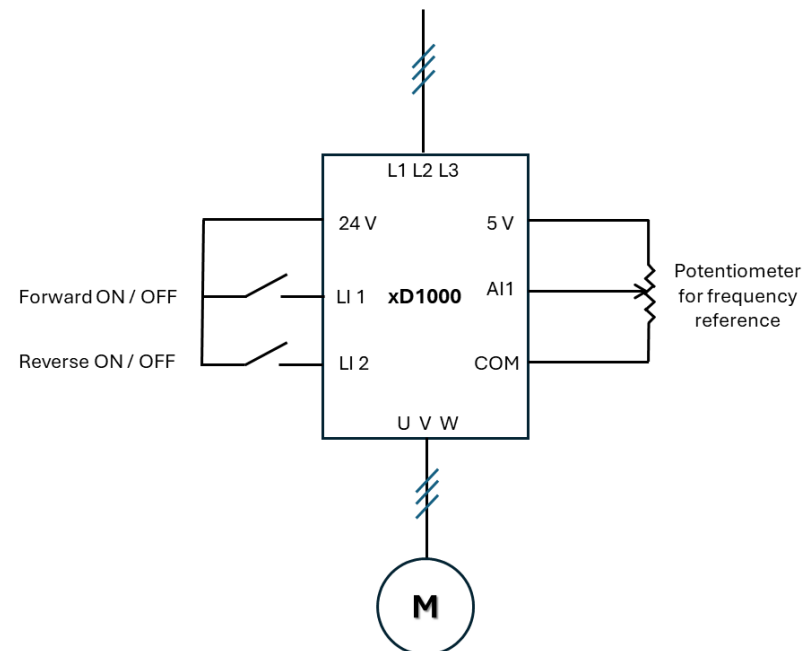
Below are parameters if start command and frequency are given from terminals (i.e. command via pushbutton & speed via potentiometer) of the drive.

First set the motor parameters according to details given on rating plate of motor.

301	Standard Motor Frequency [Select 00 for 50 Hz & 01 for 60Hz]
302	Rated Motor Power
304	Rated Motor Voltage
305	Rated Motor Current
306	Rated Motor Frequency
307	Rated Motor Speed

Parameters No. & Description	Value
401 Reference channel 1	= 01 Frequency reference from Terminal (AI1)
404 Reverse Inhibition	= 00 Reverse Direction is not inhibited.
406 Channel Configuration	= 01 Combined mode
502.0 Acceleration	= 03 sec Put the value as per application in seconds.
501.1 Deceleration	= 03 sec Put the value as per application in seconds.

Do the wiring as shown in drawing :



2. How to set motor parameters?

Parameter No.	Parameter Description	Default Value	Setting	Unit	Remark
COnF- FULL- 300- [MOTOR CONTROL MENU]					
323	Dual Rating	01 Heavy duty	00 Normal duty		up to 1.1 * I _n for 60s (For Variable Torque Application)
			01 Heavy duty		up to 1.5 * I _n for 60s (For Constant Torque Application)
301	Standard motor frequency	00 50 Hz	00 50 Hz		
302	Rated motor power	Determined by VFD rating	Set motor rated Power	kW	Enter the details as given on the motor nameplate
304	Rated motor voltage	380	Set motor rated Voltage	V	
305	Rated motor current	Determined by VFD rating	Set motor rated Current	A	
306	Rated motor frequency	50	Set motor rated Frequency	Hz	
307	Rated motor speed	Determined by VFD rating	Set motor rated Speed	RPM	
308	Maximum frequency	50	Set motor maximum Frequency	Hz	
309	Motor control type	03 Standard: 2 point V/F control without internal speed loop	03 Standard: 2 point V/F control without internal speed loop		
			00 Performance: VVC with internal speed loop		
310	IR compensation	100	100	%	Used to optimize torque at very low speed
318	Auto-tuning	00 No: use factory parameters	01 Yes: request auto-tuning		• Auto Tuning not recommended for 2 motor with 1 drive. • Auto-tuning must be performed with the motor connected and cold state. • Auto-tuning may last for 1 to 10 seconds. Do not interrupt. Wait for the display to change to 02 or 00. • Re-perform auto-tuning after motor cables are replaced to ensure effectiveness of motor control.

3. How to set acceleration / Deceleration time?

Parameter No.	Parameter Description	Default Value	Setting	Unit	Remark
CO nF- FULL- 500- 501- [Ramp menu]					
501.0	Acceleration Time	3	Set time as per application requirement	Sec	
501.1	Deceleration Time	3	Set time as per application requirement	Sec	

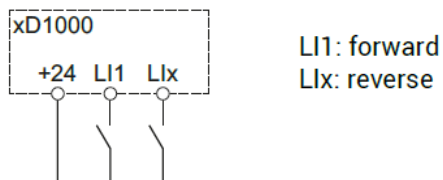
4. How to select local Start / stop button & jog dial on keypad for command and frequency reference?

Parameter No.	Parameter Description	Default Value	Setting	Unit	Remark
CO nF- FULL- 400- [Control menu]					
401	Reference channel 1	01 Analog terminal	183 Integrated display with jog Dial		Select 183 to provide frequency reference from integrated jog Dial (Available on Drive)
406	Channel configuration	01 Combined mode	01 Combined mode		Frequency channel & command channel is combined.

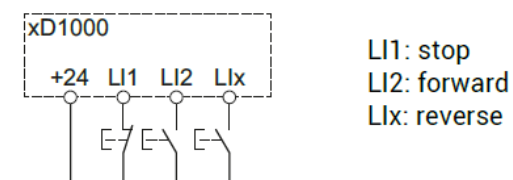
5. How to select 2 wire or 3 wire control?

Parameter No.	Parameter Description	Default Value	Setting	Unit	Remark
CO nF- FULL- 200- [I/O MENU]					
201	Type of control	00	2 Wire Control		LI 1 will be Assign to Forward & LI 2 for Reverse
		01	3 Wire Control		

2 Wire Control wiring



3 Wire Control wiring



6. How to assign digital input for reverse command?

Parameter No.	Parameter Description	Default Value	Setting	Unit	Remark
CO nF- FULL- 400- [Control menu]					
404	Reverse inhibition	00 No	00 No		Reverse direction enable
CO nF- FULL- 500- [Function menu]					
503	Reverse direction	00 Function Inactive	L2H		LI2 assigned for reverse command

7. How to select analog input for frequency reference?

Parameter No.	Parameter Description	Default Value	Setting	Unit	Remark
CO nF- FULL- 400- [Control menu]					
401	Reference channel 1	01 Analog terminal	01 Analog terminal		Select 01 to assign Analog Input 1 for frequency reference
			163 Remote Display		Select 163 to provide frequency reference from remote display (connected externally)
			164 Modbus		Select 164 to provide frequency reference from Modbus communication
			183 Integrated display with jog Dial		Select 183 to provide frequency reference from integrated jog Dial (Available on Drive)

8. How to change type of analog input from Voltage to mA?

Parameter No.	Parameter Description	Default Value	Setting	Unit	Remark
CO nF- FULL- 200- 204- [AI1 CONFIGURATION MENU]					
204.0	AI1 type	5U Voltage: 0-5 Vdc	5U Voltage: 0-5 Vdc		Select 5U for 0 to 5V input
			10U Voltage: 0-10 Vdc		Select 10U for 0 to 10V input
			0A Current: x-y mA		Select 0A for 0 to 20mA or 4 to 20mA input
204.1	AI1 current scaling of 0%	4mA	4mA	mA	Enter 4mA for 4 to 20mA & 0mA for 0 to 20mA Input
204.2	AI1 current scaling of 100%	20mA	20mA	mA	

Connect analog input at terminal AI1 & COM

9. How to assign relay or digital output for fault & Run indication?

Parameter No.	Parameter Description	Default Value	Setting	Unit	Remark
COnF- FULL- 200- 205- [R1 CONFIGURATION MENU]					
205.0	R1 assignment	01 No fault	01 No fault		Select 01 to assign R1 for fault indication & use R1C & R1A terminal of relay (NO Contact)
			02 Drive RUN		Select 02 to assign R1 for Drive RUN indication & use R1C & R1A terminal of relay (NO Contact)
205.1	R1 status (output active level)	00 Positive logic: active high	01 Negative logic: active low		Select 01 to use R1 relay for Fault indication
			00 Positive logic: active high		Select 00 to use R1 relay for Drive RUN indication
COnF- FULL- 200- 206- [LO1 CONFIGURATION MENU]					
206.0	LO1 assignment	00 Not Assigned	01 No fault		Select 01 to assign R1 for fault indication & use LO+ & LO- (only 24Vdc is allowed in LO1)
			02 Drive RUN		Select 02 to assign R1 for Drive RUN indication & use R1C & R1A terminal of relay (NO Contact)
206.1	LO1 status (output active level)	00 Positive logic: active high	01 Negative logic: active low		Select 01 to use LO1 for Fault indication
			00 Positive logic: active high		Select 00 to use LO1 for Drive RUN indication

10. How to assign analog output for output frequency or output current?

Parameter No.	Parameter Description	Default Value	Setting	Unit	Remark
COnF- FULL- 200- 216- [AO1 Configuration menu]					
216.0	AO1 assignment	00 Not Assigned	129 Output / motor current		Select 129 to show output current in Analog Output AO1
			130 Estimated motor frequency,		Select 130 to show motor frequency in Analog Output AO1
216.1	AO1 type	0A Current: 0-20 mA	10U Voltage: 0-10 Vdc		Select 10U for analog output type -> 0 to 10V
			0A Current: 0-20 mA		Select 0A for analog output type -> 0 to 20mA
			4A Current: 4-20 mA		Select 4A for analog output type -> 4 to 20mA

11. How to assign digital input for forward ON command & integrated Jog Dial for frequency reference?

Parameter No.	Parameter Description	Default Value	Setting	Unit	Remark
COnF- FULL- 400- [Control menu]					
406	Channel configuration	01 Combined mode	02 Separate mode		Frequency channel & command channel will be separated.
401	Reference channel 1	01 Analog terminal	183 Integrated display with jog Dial		Select 183 to provide frequency reference from integrated jog Dial (Available on Drive)
407	Command channel 1	01 Terminals	01 Terminals		Select 01 to give ON /OFF command from LI1
COnF- FULL- 500- [Function menu]					
503	Reverse direction	00 Function Inactive	L2H LI 2 active high		If reverse command is required, then select L2H to assign LI2 for reverse command

12. How to do the setting to start the drive automatically after Power ON?

Parameter No.	Parameter Description	Default Value	Setting	Unit	Remark
COnF- FULL- 400- [Control menu]					
406	Channel configuration	01 Combined mode	02 Separate mode		Frequency channel & command channel will be separated.
401	Reference channel 1	01 Analog terminal	183 Integrated display with jog Dial		Select 183 to provide frequency reference from integrated jog Dial (Available on Drive)
407	Command channel 1	01 Terminals	01 Terminals		Select 01 to give ON /OFF command from LI1
COnF- FULL- 200- [I/O MENU]					
202	2-wire type control	01 Transition: (Edge)	00 Level 0/1		Select 00 - Run/stop determined by level state 0/1
COnF- FULL- 600- [Fault Detection Management Menu]					
603	Catch on the fly	00 Function Inactive	01 Function Active		This function is used to enable a smooth restart if the run command is maintained after - Loss of line supply or disconnection

After this setting power OFF the drive and power ON again.

13. How to do the settings for Multispeed / Preset Speed configuration?

Preset speeds:

2, 4, or 8 speeds can be preset, requiring 1, 2 or 3 logic inputs respectively.

Here in this example, we have shown only 3 speeds which will be selected individually by LI2, LI3 & LI4 respectively. (At a time only one Logic Input will be High).

But as per application requirement you can enter more preset speeds in parameters 507.5, 507.7, 507.8 & 507.9 and use binary combination of LI to select different speeds.

See the combination table given for preset speed inputs.

8 speeds LI (507.2)	4 speeds LI (507.1)	2 speeds LI (507.0)	Speed reference
0	0	0	Preset speed 1
0	0	1	Preset speed 2
0	1	0	Preset speed 3
0	1	1	Preset speed 4
1	0	0	Preset speed 5
1	0	1	Preset speed 6
1	1	0	Preset speed 7
1	1	1	Preset speed 8

Parameter No.	Parameter Description	Default Value	Setting	Unit	Remark
COnF- FULL- 500 - 507 - [Preset speed menu]					
507.0	2 preset speeds	00 Function Inactive	L2H LI 2 active high		LI 2 is assigned to select Preset speed 2
507.1	4 preset speeds	00 Function Inactive	L3H LI 3 active high		LI 3 is assigned to select Preset speed 3
507.2	8 preset speeds	00 Function Inactive	L4H LI 4 active high		LI 4 is assigned to select Preset speed 5
507.3	Preset speed 2	10	Enter required frequency	Hz	Enter as per machine or application requirement
507.4	Preset speed 3	15	Enter required frequency	Hz	Enter as per machine or application requirement
507.6	Preset speed 5	20	Enter required frequency	Hz	Enter as per machine or application requirement

14. How to assign digital inputs (Push Buttons) for frequency UP/Down (Speed increase / Decrease) operation?

Parameter No.	Parameter Description	Default Value	Setting	Unit	Remark
COOnF- FULL- 500 - 506 - [Speed up and down]					
506.0	Up speed command	00 Function Inactive	L3H LI 3 active high		LI 3 is assigned for UP speed Push Button
506.1	Down speed command	00 Function Inactive	L4H LI 4 active high		LI 4 is assigned for Down speed Push Button
506.2	Store	00 Do not save	00 Do not save		Select 00 if you don't want to save last frequency reference. After stop command frequency reference return to 0.
			01 Save in RAM		Select 01 to save last frequency reference in RAM. After power off the drive frequency reference returns to 0.
			02 Save in ROM		Select 02 to save last frequency reference in ROM. After stop or power off frequency reference will not return to 0

15. How to clear the --06 Error code in xD1000 Drive?

Description : --06 stands for NST, it is not a fault but it means the drive is in an freewheel stop condition.

Causes and fixes :

1. VFD has an input assigned to Freewheel stop? Check to see if that the input is active.
2. The drive is in 3 wire control (201 = 3C). LI1 must be active before the NST will clear.
3. The VFD is in 2 wire control (201 = 2C) Verify if the drive was powered with the run command active, if so cycle the run command (means first give OFF command & again give ON command) or set 202 to 01 (LEVEL) IO menu, 2 wire type = level. (For power ON RUN function)
4. If the drive is being controlled by serial communication, the drive will display --06 until the Hex value of 6 has been written to the command word.
5. If the 4-20 mA signal loss is programmed to freewheel stop and the drive has loss the signal. Check the 4-20 mA signal.
6. If the drive is running from terminal control and the stop button on the keypad or remote display is pressed, the drive will go to --06. Power cycle the drive to clear the --06.

16. Why F015 fault occurs in xD1000 Drive?

Name of fault :- 3 Output phases loss

Possible causes :-

- Motor not connected
- Motor power too low, below 6% of the rated drive current
- Output contactor open
- Transient instability in the motor current

Remedy :-

- Check the connections from the drive to the motor.
- To Test drive operation with low power motor or without a motor.

In factory settings mode, motor phase loss detection is active Output Phase loss detection i.e 605 = 01.

To check the drive in a test or maintenance environment, without having to use a motor with the same rating, motor phase loss detection or Output Phase loss detection needs to be deactivated i.e 605 = 00 & again set it to 1 when connecting back the motor.

- Check and optimize IR compensation in parameter 310, Rated motor voltage in parameter 304 and Rated motor voltage in parameter 305 and then perform Auto-tuning by setting parameter 318.

17. How to troubleshoot **F018** fault in xD1000 drive?

Name of Fault : Motor short circuit fault.

Possible causes are listed below:

1. Short circuit or grounding at drive output.
2. Ground fault in run state.
3. Significant current leakage to ground, while several motors are connected in parallel.

Resolution:

1. Using the megger test check the motor cable and motor insulation.
2. Try connecting the motor directly to the line power & observe the current flow at minimum load. It should be within the range on the motor name plate.
3. Disconnect the motor from the drive and see if the fault is still occurring. If the fault is still present, then the issue could be with the drive.
4. Perform an IGBT test to see if the issue is with any IGBT's of the drive. Refer to page 107 of the xD1000 programming manual. If an IGBT is in fault state, it will be shown after the test.

18. Why does **F010** fault occurs in xD1000 Drive?

Name of fault : Over current

Possible causes:

1. Parameters in the Motor control menu are not correct.
2. Inertia or load too high
3. Mechanical locking

Remedy :

- Check the motor control parameters.
- Check the size of the motor/drive/load.
- Check the state of the mechanism.
- Connect line motor chokes.
- Reduce the Switching frequency range.
- Check the ground connection of drive, motor cable and motor insulation.

19. How do you set up the xD1000 for auto - manual or local - remote control?

Requirement: In Manual mode, frequency reference is from UP / DOWN push buttons and in Auto mode, frequency reference is from Analog input 1.

In Auto as well as in Manual mode, ON/OFF command is from Terminal LI 1

When Auto/Manual selector switch is open [Low / 0] then Manual mode will be selected & when switch is closed [High / 1] then Auto mode will be selected

Parameters No. & Description	Value	
301 Standard Motor Frequency	=	Select 00 for 50 Hz & 01 for 60Hz
302 Rated Motor Power		
304 Rated Motor Voltage		
305 Rated Motor Current		
306 Rated Motor Frequency		
307 Rated Motor Speed		
502.0 Acceleration	=	03 sec Put the value as per application in seconds.
501.1 Deceleration	=	03 sec Put the value as per application in seconds.
401 Reference channel 1	=	163 Remote Display
406 Channel Configuration	=	02 Separate mode
407 Command channel 1	=	01 Terminal LI 1 is assigned for On/Off command in Manual mode
408 Forced Local Assignment	=	L2H LI2 is assigned for Auto Manual selection
409 Forced Local Frequency Reference	=	01 Frequency reference from analog input 1 in Auto Mode
410 Forced Local On/Off command	=	01 On/Off command from Terminal LI 1 in Auto Mode
506.0 Up speed command	=	L3H LI3 is assigned for speed increase command
506.1 Up speed command	=	L4H LI4 is assigned for speed decrease command
506.2 Speed Store	=	02 Select 02 to save last frequency reference in ROM.

Wiring Diagram –

